GRouNdScale TF-Count Scaling Report

Auto-generated from `publication/tf-num-experiment` run logs

October 16, 2026

1 Overview

This report analyses the 14 completed runs of the publication TF-count experiment. The number of TFs was swept from 200 to 2,800 in steps of 200 while the GRN parameter k was held fixed at 15. Results are indexed by the number of *target genes* in the causal graph, as requested. Logs are taken from Slurm job `35274208`, which matches the current `results/` directory. The older sweep (job `33898186`/`33898382`, sizes 100 to 2,900) and `results-old/` are excluded.

Requested versus realised TFs. The sweep requests a number of TFs, but the causal graph that is actually trained on retains a different number once the bipartite filtering removes TFs that regulate only other selected TFs. The realised count matches the request up to 1,200 TFs, is nearly equal at 1,400 and 1,600, and then falls below it (1,155 realised for 2,800 requested). This report uses the *realised* counts from the causal-graph block of each log, and Table 1 lists both.

Genes discarded and the fit subset. The dataset has 3,000 genes. As the TF count grows, some genes are discarded from the causal graph; the number discarded is listed in Table 1. Runs with fewer than 100 genes discarded (requested TFs 200–1800, 9 runs) form a *fit subset*. Beyond that, discarding grows quickly, so the subset fit tests whether the discarding is what drives the scaling.

Missing run. There is no 3,000-TF run in either `results/` or the logs. With 3,000 genes in the dataset, 3,000 TFs would leave no target genes, which is the likely reason it was not formed; no log confirms this directly.

2 GRN Characteristics

Table 1: GRN statistics per run, ordered by target genes. *Requested* is the swept TF count; the remaining columns are read from the causal-graph block of each log. Genes discarded is 3,000 minus the genes retained in the causal graph.

Target genes	Requested TFs	TFs	Genes	Genes discarded	TF/Target (%)	Imposed Edges	Possible Edges	Density (%)
200	2800	1155	1355	1645	577.50	3,000	231,000	1.3
400	2600	1577	1977	1023	394.25	6,000	630,800	1.0
600	2400	1782	2382	618	297.00	9,000	1,069,200	0.8
800	2200	1873	2673	327	234.12	12,000	1,498,400	0.8
1000	2000	1844	2844	156	184.40	15,000	1,844,000	0.8
1200	1800	1739	2939	61	144.92	18,000	2,086,800	0.9
1400	1600	1586	2986	14	113.29	21,000	2,220,400	0.9
1600	1400	1399	2999	1	87.44	24,000	2,238,400	1.1
1800	1200	1200	3000	0	66.67	27,000	2,160,000	1.2
2000	1000	1000	3000	0	50.00	30,000	2,000,000	1.5
2200	800	800	3000	0	36.36	33,000	1,760,000	1.9
2400	600	600	3000	0	25.00	35,986	1,440,000	2.5
2600	400	400	3000	0	15.38	38,934	1,040,000	3.7
2800	200	200	3000	0	7.14	41,610	560,000	7.4

3 Training Step Times

Table 2: Average final-step times (ms) for the causal controller (first phase) and causal generator (second phase).

Target genes	Requested TFs	Controller (ms)	Generator (ms)
200	2800	36	45
400	2600	34	67
600	2400	35	74
800	2200	37	81
1000	2000	36	100
1200	1800	38	104
1400	1600	39	124
1600	1400	36	126
1800	1200	36	152
2000	1000	18	144
2200	800	30	173
2400	600	35	193
2600	400	37	202
2800	200	40	211

Table 3: Expected duration for 200,000 controller steps and 500,000 generator steps.

Target genes	Requested TFs	Controller	Generator	Total
200	2800	2.0 h	6.2 h	8.2 h (0.34 d)
400	2600	1.9 h	9.3 h	11.2 h (0.47 d)
600	2400	1.9 h	10.3 h	12.2 h (0.51 d)
800	2200	2.1 h	11.2 h	13.3 h (0.55 d)
1000	2000	2.0 h	13.9 h	15.9 h (0.66 d)
1200	1800	2.1 h	14.4 h	16.6 h (0.69 d)
1400	1600	2.2 h	17.2 h	19.4 h (0.81 d)
1600	1400	2.0 h	17.5 h	19.5 h (0.81 d)
1800	1200	2.0 h	21.1 h	23.1 h (0.96 d)
2000	1000	1.0 h	20.0 h	21.0 h (0.88 d)
2200	800	1.7 h	24.0 h	25.7 h (1.07 d)
2400	600	1.9 h	26.8 h	28.8 h (1.20 d)
2600	400	2.1 h	28.1 h	30.1 h (1.25 d)
2800	200	2.2 h	29.3 h	31.5 h (1.31 d)

4 Parameter Counts

Target genes	Generator (excl. CC)	Controller	Critic	Labeler	Anti-Lab.	Gen. Opt.	Crit. Opt.	Lab. Opt.	Anti-Lab. Opt.
200	0.52M	2.09M	2.04M	10.74M	10.74M	0.56M	6.13M	32.19M	32.19M
400	1.04M	2.72M	2.68M	11.99M	11.99M	1.11M	8.05M	35.92M	35.92M
600	1.56M	3.14M	3.10M	12.80M	12.80M	1.67M	9.29M	38.35M	38.35M
800	2.07M	3.44M	3.39M	13.38M	13.38M	2.23M	10.18M	40.10M	40.10M
1000	2.59M	3.61M	3.57M	13.72M	13.72M	2.79M	10.71M	41.12M	41.12M
1200	3.11M	3.71M	3.67M	13.91M	13.91M	3.34M	11.00M	41.69M	41.69M
1400	3.63M	3.76M	3.72M	14.00M	14.00M	3.90M	11.15M	41.97M	41.97M
1600	4.15M	3.77M	3.73M	14.03M	14.03M	4.46M	11.19M	42.05M	42.05M
1800	4.67M	3.77M	3.73M	14.03M	14.03M	5.02M	11.19M	42.06M	42.06M
2000	5.19M	3.77M	3.73M	14.03M	14.03M	5.57M	11.19M	42.06M	42.06M
2200	5.70M	3.77M	3.73M	14.03M	14.03M	6.13M	11.19M	42.06M	42.06M
2400	6.22M	3.77M	3.73M	14.03M	14.03M	6.68M	11.19M	42.06M	42.06M
2600	6.72M	3.77M	3.73M	14.03M	14.03M	7.23M	11.19M	42.06M	42.06M
2800	7.16M	3.77M	3.73M	14.03M	14.03M	7.69M	11.19M	42.05M	42.05M

Table 4: Parameter counts for each component and optimiser, from `params-size.json`. The generator column excludes the causal controller, which is a submodule of the stored `gen_state_dict` and is shown separately.

5 GPU Memory

Table 5: Peak GPU memory (MiB) by training phase, from the monitoring CSV using the logged phase windows.

Target genes	Requested TFs	Controller (MiB)	Generator (MiB)
200	2800	6,075	7,261
400	2600	6,185	7,519
600	2400	6,199	7,661
800	2200	6,147	7,829
1000	2000	6,207	7,965
1200	1800	6,233	8,065
1400	1600	6,231	8,197
1600	1400	6,231	8,337
1800	1200	6,207	8,421
2000	1000	6,159	8,415
2200	800	6,159	8,559
2400	600	6,207	8,681
2600	400	6,159	8,843
2800	200	6,159	8,959

6 Linear Fits in Target Genes

Let T denote the number of target genes. Two ordinary least-squares lines are reported for each quantity: one fitted to all 14 runs, and one fitted to the fit subset (fewer than 100 genes discarded). Quadratic fits are not reported, since the theoretical model does not support curvature in these quantities.

$$\widehat{P}_{\text{gen}}(T) = 2576\,T + 1.719 \times 10^4 \text{ parameters}, \quad R^2 = 0.9999 \quad \text{(all runs)}, \tag{1}$$

$$\widehat{P}^{\text{sub}}_{\text{gen}}(T) = 2553\,T + 6.587 \times 10^4 \text{ parameters}, \quad R^2 = 0.9996 \quad \text{(fit subset)}, \tag{2}$$

$$\widehat{M}_{\text{gen}}(T) = 0.5976\,T + 7297 \text{ MiB}, \quad R^2 = 0.9819 \quad \text{(all runs)}, \tag{3}$$

$$\widehat{M}^{\text{sub}}_{\text{gen}}(T) = 0.5283\,T + 7441 \text{ MiB}, \quad R^2 = 0.9772 \quad \text{(fit subset)}, \tag{4}$$

$$\widehat{t}_{\text{gen}}(T) = 0.06290\,T + 33.93 \text{ ms/step}, \quad R^2 = 0.9848 \quad \text{(all runs)}, \tag{5}$$

$$\widehat{t}^{\text{sub}}_{\text{gen}}(T) = 0.06808\,T + 22.61 \text{ ms/step}, \quad R^2 = 0.9659 \quad \text{(fit subset)}. \tag{6}$$

Table 6: Goodness of fit in target genes, for all runs and for the fit subset.

Quantity	R^2 (all)	RMSE (all)	R^2 (subset)	RMSE (subset)
Generator parameters	0.9999	22,937.49	0.9996	25,125.15
Generator peak memory (MiB)	0.9819	65.35	0.9772	41.71
Generator step time (ms)	0.9848	6.30	0.9659	6.61

7 Graphs

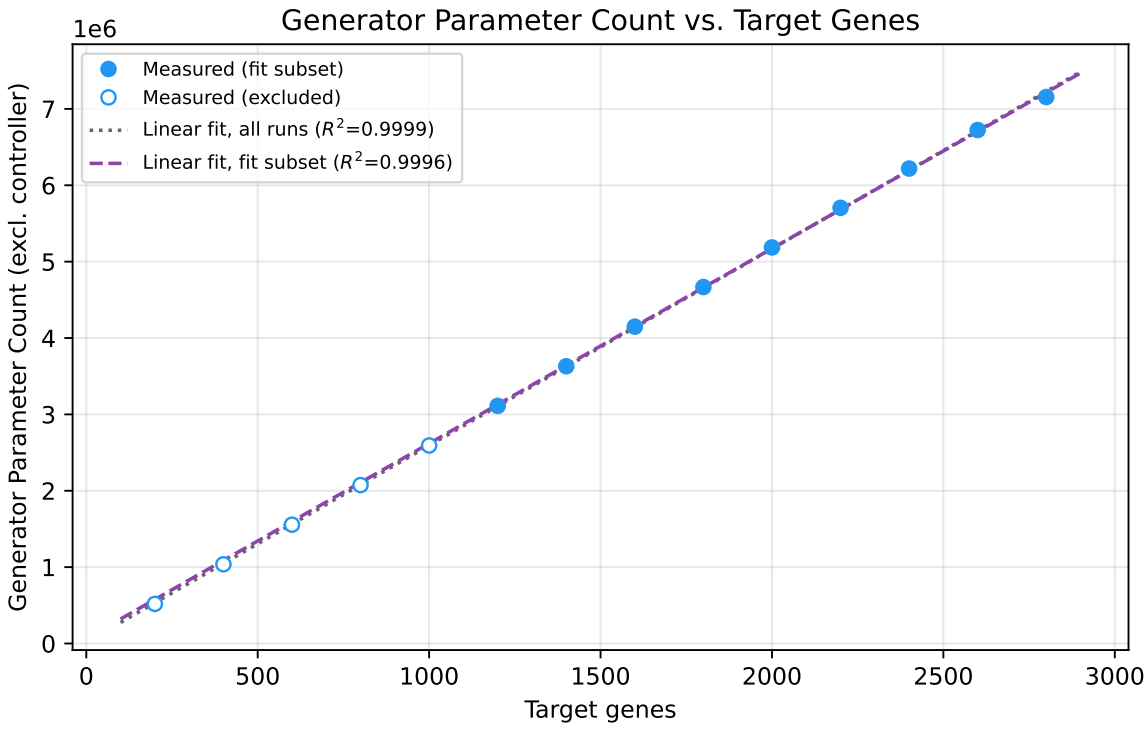


Figure 1: Generator parameter count (excluding the controller) versus target genes. Filled points are the fit subset; hollow points are excluded runs. Dotted: linear fit to all runs; dashed: linear fit to the subset.

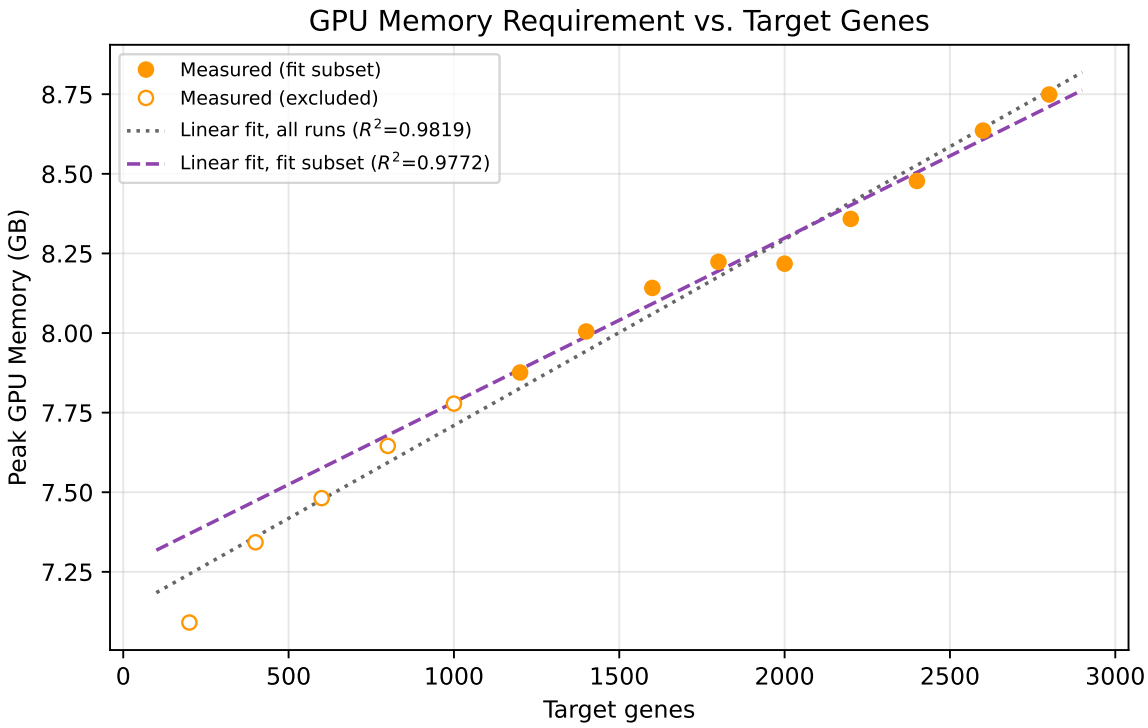


Figure 2: Peak generator GPU memory versus target genes, with the same two linear fits.

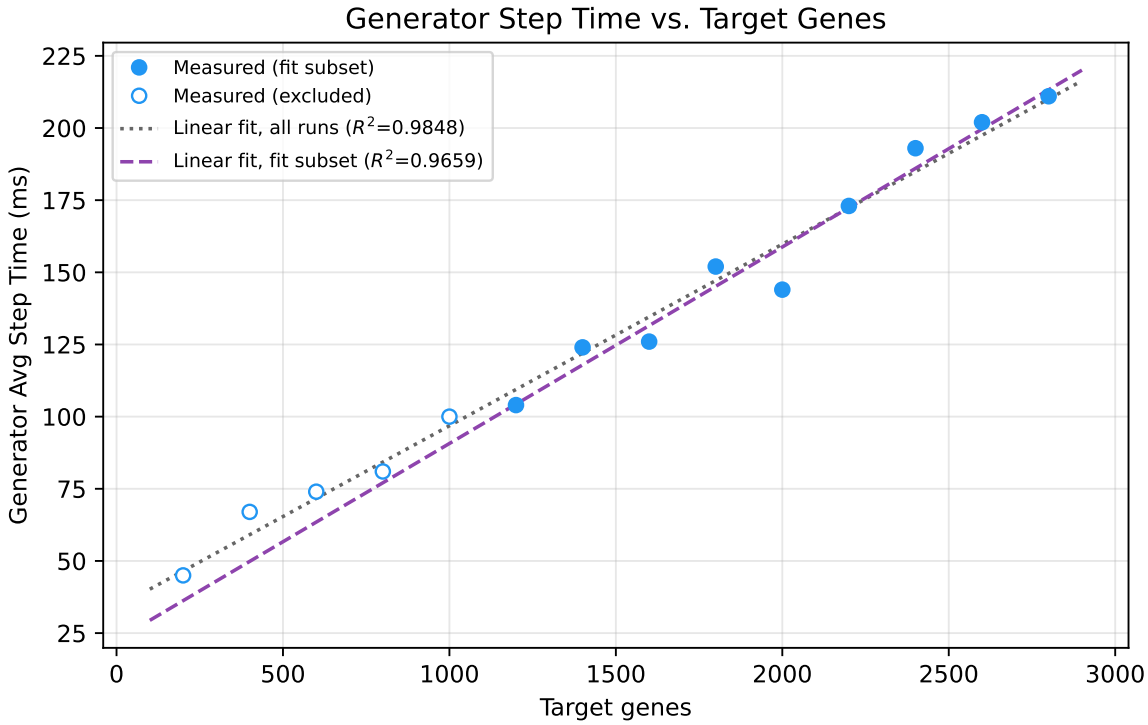


Figure 3: Causal-generator average step time versus target genes, with the same two linear fits.

8 Notes

- Fit subset: the 9 runs with fewer than 100 genes discarded (largest discard in the subset: 61; smallest discard among excluded runs: 156). Comparing the subset line with the all-run line: parameters: slope -0.9%, subset line misses excluded runs by up to 11.2%; peak memory: slope -11.6%, subset line misses excluded runs by up to 3.9%; step time: slope +8.2%, subset line misses excluded runs by up to 25.6%. The subset covers a narrower target range (1,200 to 2,800), so these slope changes carry more uncertainty than the R^2 values.

- Scope of each input. Generator parameters are `gen_state_dict` minus the controller's parameters, so they describe the causal generator alone. Optimiser counts are already generator-only: the generator optimiser holds about three times the sparse-generator parameter count, far below what it would be if the controller were included. Peak GPU memory is the maximum inside the generator-training window, which starts after the controller window closes. The controller's resident weights are about 14 MiB, under 0.2% of the generator peak. Step time is the generator-phase step logged by the trainer; the causal generator's forward pass calls the controller, so that time is not separable from the logs.

- TF/Target exceeds 100% from 1,600 requested TFs onward, meaning the graph has more TFs than target genes; the ratio is reported as a plain percentage for consistency with the other experiments.

- Imposed edges fall from 41,610 to 3,000 across the sweep, and the generator's parameter count falls with them. Target genes therefore track the sparse-edge budget as well as the output dimension, so the fits in T describe this sweep rather than a pure dependence on target count.

- Runs with an out-of-memory error: 0 of 14.